

Course Outline

(Way to Move)

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Course Title: Data Structure
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Introduction

- ① **Course Title:** Data Structure
- ② **Course No. :** CSE 1201
- ③ **Contact hours/week:** 3
- ④ **Credits:** 3.00

Course Description

- 1 Introduction
- 2 Algorithm and Complexity
- 3 Linear Array
- 4 Stack
- 5 Queue
- 6 Linked List (Array and Pointer)
- 7 Tree
- 8 Sorting Algorithm

Grading

- 1 Quizzes/Class Test: **20 Marks*** (3 best out of 4 quizzes/class test may be taken for awarding grade)
- 2 Homeworks/Attendance: **8 Marks***
- 3 Semester Exam: **72 Marks**

*** - We reserve the right to change the above grading scheme.**

Homeworks

- A new homework is released and is then due after 9 days. No grade will be given to homework submitted afterwards.
- Homework solutions should be written and submitted individually, but **discussions among students are encouraged.**

Course Outline

- **Cycle:** 1st and 2nd
- **Total Class:** 6
- **Topics:**
 - **Introduction:** Concepts and Examples of Elementary Data Objects, Necessity of Structured Data, Types of Data Structure, Ideas on Linear and Nonlinear Data Structure.
 - **Complexity:** Algorithm and flow chart, Complexity of algorithms, Rate of growth, Big-O notation, Complexity of Linear Search, Binary search & Bubble sort algorithm.

Course Outline

- **Cycle:** 3rd and 4th(2)
- **Total Class:** 5
- **Topics:**
 - **Linear Array:** Linear Array & its representation in memory, Traversing LA, Insertion & Deletion in LA, Bubble Sort, Linear Search & binary Search, Multidimensional Array & its representation in memory, Algebra of matrices, Sparse matrices.

Course Outline

- **Cycle:** 4th(1),5th and 6th(2)
- **Total Class:** 6
- **Topics:**
 - **Linked List:** Linked list & its representation in memory, Traversing, Searching, Insertion & Deletion operation on Linked list, Circular List, Header linked lists, Two way lists.

Course Outline

- **Cycle:** 6th(1),7th and 8th(1)
- **Total Class:** 5
- **Topics:**
 - **Stack:** Stack representation & applications; PUSH and POP operation on stack. Polish Notation, reverse polish notation; Evaluation of a postfix expression; Transforming infix expression into postfix expression.

Course Outline

- **Cycle:** 8th(2) and 9th
- **Total Class:** 5
- **Topics:**
 - **Queue:** Its representation, Insertion & deletion in Queue, Priority Queues, Recursion [Factorial function, Fibonacci sequence, Ackermann function, Towers of Hanoi].

Course Outline

- **Cycle:** 10th and 11th
- **Total Class:** 6
- **Topics:**
 - **Tree:** Tree terminology, representation of binary trees in memory, Traversing binary tree, Binary search tree, Insertion & deletion on binary search tree, Insertion & deletion on heap, Heap sort, B trees, General tree.

Course Outline

- **Cycle:** 12th
- **Total Class:** 3
- **Topics:**
 - **Sorting:** Insertion sort, selection sort, quick sort, merge sort, Searching & data modification, Hash function, collision resolution, Chaining.

Referred Books



Seymour Lipschetz

Data Structure



Y. Langsam, Augenstein, A. M. Tanenbaum

Data Structures Using C and C++



Edward M. Reingold

Data structures

CSE 1202: Sessional Based on CSE 1201

Introduction

- ① **Course Title:** Sessional Based on CSE 1201
- ② **Course No. :** CSE 1202
- ③ **Contact hours/week:** 3.00
- ④ **Credits:** 1.5

CSE 1202: Sessional Based on CSE 1201

Modules

Module	Topics	Cycle
01	Complexity of Algorithms	01 and 02
02 and 03	Linear Array (Arrays, Records and Pointers)	03 and 04
04 and 05	Linked List	05 and 06
06	Stack	07 and 08
07	Queue	08 and 09
08	Tree	10 and 11
09	Sorting	12

* **Seymour Lipschetz** : Data Structure .

Tools

1 C

Grading

- 1 **Quiz Test:** 20 Marks*
- 2 **Homeworks/Attendance:** 8 Marks*
- 3 **Board Viva:** 25 Marks
- 4 **Others:** 47 Marks*

	Marks Distribution
Lab Report (Individual)*	20
Lab Performance(Individual)*	27
Example: Average Performance(Every Lab) 17 Viva 10 (Average)	
Total	47

* - We reserve the right to change the above grading scheme.

In Lab

- 1 Lab Report
- 2 Books, Papers and Pens
- 3 Laptop* **(if individual PC/LAPTOP is not available)**

Referred Books



Seymour Lipschetz

Data Structure



Y. Langsam, Augenstein, A. M. Tanenbaum

Data Structures Using C and C++



Edward M. Reingold

Data structures

Web: Short URL

① <https://sites.google.com/site/t062000/ds18>

The End